



Xi'an Jiaotong-Liverpool University

西交利物浦大學

FIN302

Finance and Markets

Lecture 9: Swaps

Prof Tony He

Lecture 9- Swaps

Questions and Learning Outcomes

- What are the swaps, what for, and how do they work?
 - Interest rate, currency, and index swaps
 - Comparative advantages
 - Design and value of swaps
- What are the risks involved in swaps?
 - Credit and liquidity risks

Lecture 9- Swaps

- Readings:
 - Hull (2022): Chapters 7
 - Additional reading Chapter 25
- Tutorial Problems from Hull:
 - 7.11-16, 19, 21.

Swap---Basics

➤ Swap:

- an agreement between two counter parties to exchange a sequence of cash flows
 - **Interest Rate (IR) Swaps**
 - **Foreign Exchange Rate (FX) Swaps**
- structured like bond cash flows
- maturities extend from 1 to 30 years or more

➤ Size of the swap market

- Growth since their introduction in 1980 has been staggering
- 2001: \$51 trillion (IR swaps) and \$4 trillion (FX swaps)
- 2010: \$347 trillion (IR swaps) and \$8 trillion (FX swaps)

Swap---What for?

Basic aspects of currency swaps and interest rate swaps:

- **Manage interest rate and currency risks**
- Allow companies to **change**
 - the currency of denomination and/or
 - the nature of the interest rate on their debts
- **Minimizing the cost of debt** - shaving off even a few basis points can save a company millions of dollars
- **Manipulating earnings** – swaps can be used by managers to ensure that they meet earnings expectations

Interest Rate (IR) Swaps

- The features of **interest rate swaps**:
 - No initial exchange of principals
 - **Netting** of interest payments
 - The cash flows are determined in advance but paid in arrears

- It is used to
 - Convert a **liability** from a fixed rate to a floating rate or vice versa
 - Convert an **investment** from a fixed rate to a floating rate or vice versa

Plan Vanilla Interest Rate Swaps

- **Example:** Consider an agreement by Microsoft to **receive 6-month LIBOR** and **pay a fixed rate of 5%** per annum, every 6 months for 3 years, on a notional principal of \$100 million
- Fixed cash flow paid = $\$100 \text{ m} \times 0.05 \times 1/2 = \2.50 million
 - Current 6-month LIBOR = 4.2%
 - Floating cash flow received = $\$100\text{m} \times 0.042 \times 1/2 = \2.10 million
 - In 6-months, 6-month LIBOR = 4.8%
 - Floating cash flow received = $\$100 \text{ m} \times 0.048 \times 1/2 = \2.40 million

Date	LIBOR Rate	FLOATING Cash Flow	FIXED Cash Flow	Net Cash Flow
Mar.5, 2001	4.2%			
Sept. 5, 2001	4.8%	+2.10	-2.50	-0.40
Mar.5, 2002	5.3%	+2.40	-2.50	-0.10
Sept. 5, 2002	5.5%	+2.65	-2.50	+0.15
Mar.5, 2003	5.6%	+2.75	-2.50	+0.25
Sept. 5, 2003	5.9%	+2.80	-2.50	+0.30
Mar.5, 2004	6.4%	+2.95	-2.50	+0.45

Interest Rate Swaps: Absolute Advantage

Classical swap example: The following rates are offered to two parties

	<i>Fixed</i>	<i>Floating</i>
AAACorp	4.00%	6-month LIBOR - 0.10%
BBBCorp	5.20%	6-month LIBOR + 0.60%

➤ Absolute advantage

- AAACorp is always better than BBBCorp by 1.2% and 0.7% in borrows at fixed and floating rates, respectively.
- This may reflect their difference in credit rating---high rating means low borrowing costs.

Interest Rate Swaps: Nature and Needs

➤ Company AAA

- Cash flows (CFs) are pro-cyclical---good (bad) CFs when economy is good (bad)
- To manage CFs, **AAA wants to borrow at short-term (floating) rate** when interest rate is also pro-cyclical---pay high (low) interest rate when CFs are high (low)

➤ Company BBB

- Cash flows (CFs) are very stable, and not affected by the economic cycle
- To manage CFs, **BBB wants to borrow at a fixed rate** for the long term.

➤ **Question:** Is it possible to reduce

- the *floating borrowing rates* for AAA and
- the *fixed borrowing rate* for BBB?



Interest Rate Swaps: Comparative Advantages

➤ Comparative advantages:

- The differences between AAA and BBB:
 - Fixed rate: 1.2%
 - Floating rate: 0.7%
- Relatively,
 - ***AAA is better at borrowing at a fixed rate***
 - ***BBB is better at borrowing at a floating rate***

➤ Swap:

- *AAA can borrow at a fixed rate for BBB*
- *BBB can borrow at a floating rate for AAA*
- Effectively,
 - AAA converts a fixed rate to a more attractive floating rate
 - BBB converts a floating rate to a more attractive fixed rate

Interest Rate Swaps: Comparative Advantages

- Without the swap:

AAA	pays	LIBOR-0.10%
BBB	pays	5.20%
Total =		LIBOR + 5.10%

- With the swap:

AAA	pays	4%
BBB	pays	LIBOR + 0.60%
Total =		LIBOR + 4.60%

- The **total gain** from the swap is
 $(\text{LIBOR} + 5.10\%) - (\text{LIBOR} + 4.60\%) = 0.50\%$
- The **effective borrowing rate if split equally**:
 - AAA**: floating rate at $(\text{LIBOR} - 0.1\%) - 0.25\% = \text{LIBOR} - 0.35\%$
 - BBB**: the fixed rate at $5.2\% - 0.25\% = 4.95\%$

Swap Market Maker: Financial Institutions (FIs)

- **Financial Institutions (FIs)** can act as market makers for swaps
 - Since it is unlikely that two parties desiring offset a swap will approach a FI at the same time
 - Most swaps are entered into by the FI without an offsetting position
 - The FI requires a percentage of the swap value as a commission
- FI can hedge the counterparty risk
 - **Market risk**, for instance, interest rate risk - hedged by entering offsetting contracts
 - **Default risk** - hedged in the credit risk market

Interest Rate Swaps: Market Maker

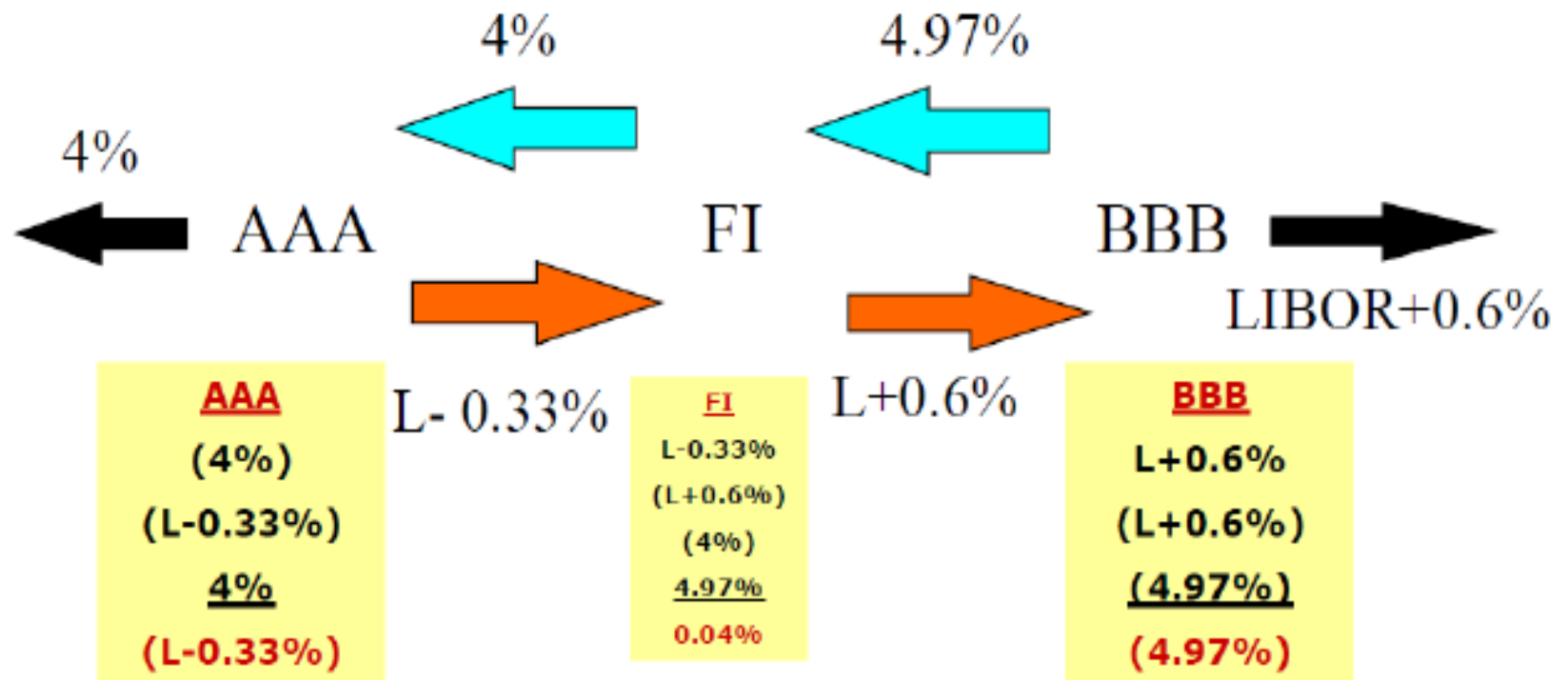
Suppose a Financial Institution (Market Maker) requires 4 basis points as commission

- Each party will benefit by
 $(0.50\% - 0.04\%) / 2 = 0.23\%$
- The **effective borrowing rate**:
 - **AAA: LIBOR-0.33%**
 - **BBB: 4.97%**
- Both parties have improved their borrowing rate by 0.23%. This is the comparative advantage of the swap

Interest Rate Swaps: Design

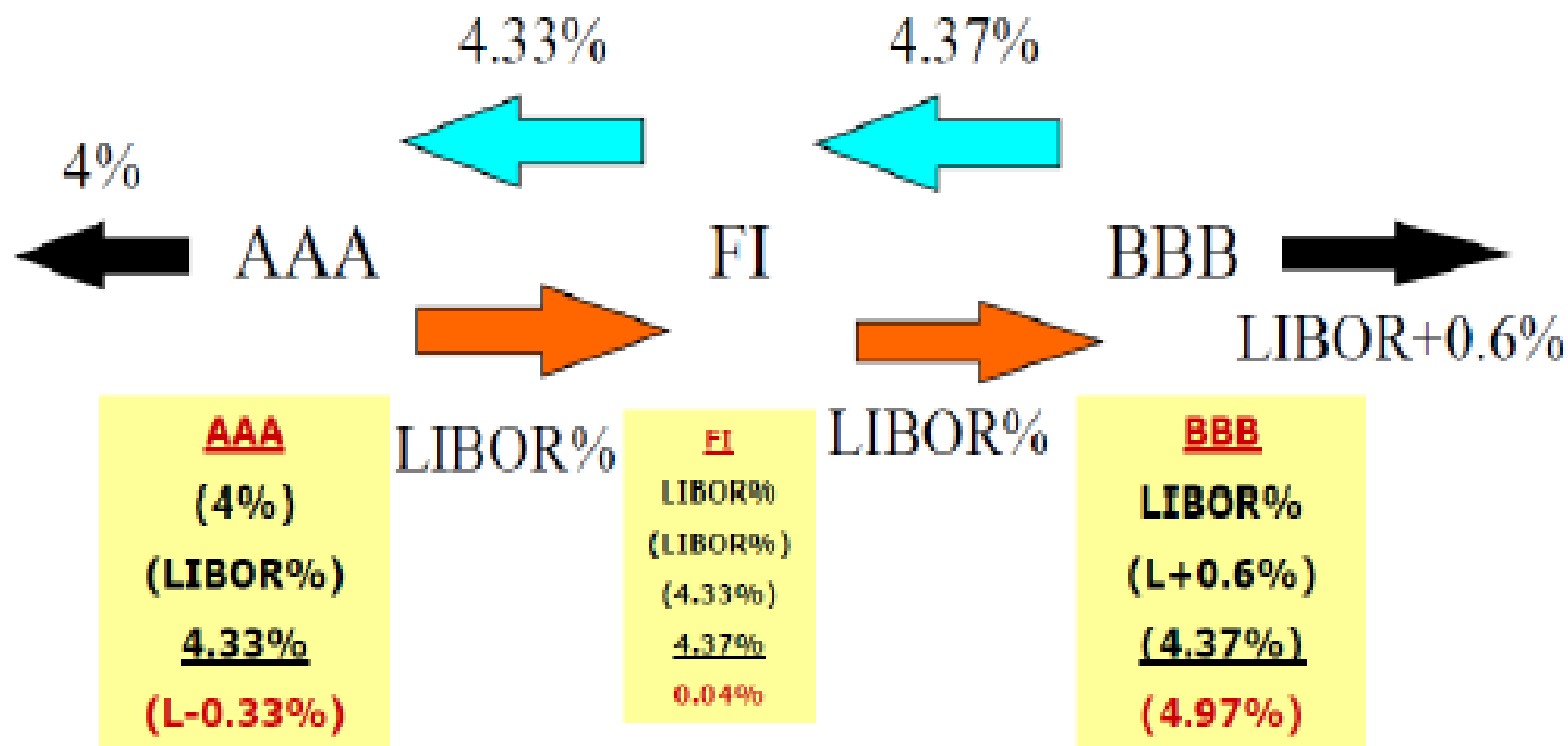
Interest Rate Swap Design

- Step 1: AAA borrows at 4% and BBB borrows at $L+0.60\%$ from outside lenders
- Step 2: AAA receives 4% and BBB receives $L+0.60\%$ from FI
- Step 3: AAA pays $L-0.10\%-0.23\% = L-0.33\%$ and BBB pays $(5.20-0.23)\% = 4.97\%$ to the FI



Design of Interest Rate Swaps

➤ Example: Alternative designs



Interest Rate Swaps: Some Issues

- Criticism of the comparative advantage arguments
 - The fixed rates are for 5 years, while the LIBOR rates are for six months. These rates are reviewed every six months and the lender has the option of changing the spread over LIBOR
 - Therefore BBBCorp's **fixed rate** depends on the spread above LIBOR charged in the future, which **may not be "fixed"**

Interest Rate Swaps: Some Issues

- Changed circumstances may change the preference for the nature of debt
 - ***Floating rate:***
 - A company whose CFs are pro-cyclical can borrow at short-term rates which are also pro-cyclical, but short-term rates are not without risk though – lenders can just cut off funds to borrowers
 - Default risk increases when the floating rate increases more than the fixed rate.
 - ***Fixed rate:***
 - A company whose CFs are stable can borrow at a long-term fixed rate, which may face the risk of default in economic contractions
 - Default risk increases when the fixed rate is higher

Value of Interest Rate Swaps

- Interest rate swaps can be valued as the difference between the values of a fixed-rate bond and a floating-rate bond
 - The ***fixed-rate bond*** is valued in the usual way
 - The **floating rate bond is valued by assuming that it is worth par** immediately after the next payment date
- The value of an interest rate swap where floating cash flows are received and fixed cash flows are paid

$$V_{swap} = B_{fl} - B_{fx}$$

- Alternatively, an interest rate swap can be valued as a ***portfolio of forward rate agreements*** (FRAs)

Value of Interest Rate Swaps

- **Example:** Notional \$100 million with 10 months remaining
- Receive 6-month LIBOR and pay 7% semi-annually
 - Swap discount rate is 5% p.a. (continuously compounded) for all maturities
 - Latest LIBOR is 4.6%
 - The floating rate bond is worth par immediately after an interest payment
- **Outcom:**
- Fixed payments = \$100 m x 7% x 6/12 = \$3.5 m
$$B_{fx} = 3.5 e^{-0.05 \frac{4}{12}} + 103.5 e^{-0.05 \frac{10}{12}} = \$102.718 \text{ m}$$
 - Receives floating payment = \$100 m x 4.6% x 6/12 = \$2.3 million
$$B_{fl} = 102.3 e^{-0.05 \frac{4}{12}} = \$100.609 \text{ m}$$
 - Value of the swap to the party receiving floating
$$V_{swap} = B_{fl} - B_{fx} = -\$2.109 \text{ million}$$
leading to potential default!

Currency Swaps

- The features of currency swaps are
 - ***The principal is exchanged*** at the beginning and the end of the swap
 - All interest payments are exchanged in full in a different currency
 - Recall that in an interest rate swap, the principal is not exchanged and interest payments are netted
- It is used to convert a liability (investment) from one currency to another currency

Currency Swaps

- **Example:** IBM wants to borrow sterling using USD. It has agreed to **pay 11% on a sterling** principal of £10 million and **receive 8% on a USD** principal of \$15 million every year for 5 years

	Dollars \$	Pounds £
Year	-----millions-----	
2001	-15.00	+10.00
2002	+1.20	-1.10
2003	+1.20	-1.10
2004	+1.20	-1.10
2005	+1.20	-1.10
2006	+16.20	-11.10



Currency Swaps: Example

Example: The following rates are offered to two companies

	USD	AUD
General Motors	5.0%	7.6%
Qantas	7.0%	8.0%

- GM: a US company that wants to borrow AUD
- Qantas: an Aussie company that wants to borrow USD
- Notice the spread difference
 - GM has a comparative advantage in USD rates
 - Qantas has a comparative advantage in AUD rates
- A currency swap will allow
 - GM to access the comparatively better AUD rate
 - Qantas to gain access to a comparatively better USD rate

Currency Swaps: Comparative Advantages

- **Comparative advantage** argument for why both companies should enter into the currency swap

- Without the swap:

GM	pays on a AUD loan at 7.60%
Qantas	pays on a USD loan at 7%
Total	= 14.60%

- With the swap:

GM	pays on a USD loan 5%
Qantas	pays on a AUD loan 8%
Total	= 13%

- The total gain from the swap: $14.60\% - 13\% = 1.60\%$

Currency Swaps : Comparative Advantages

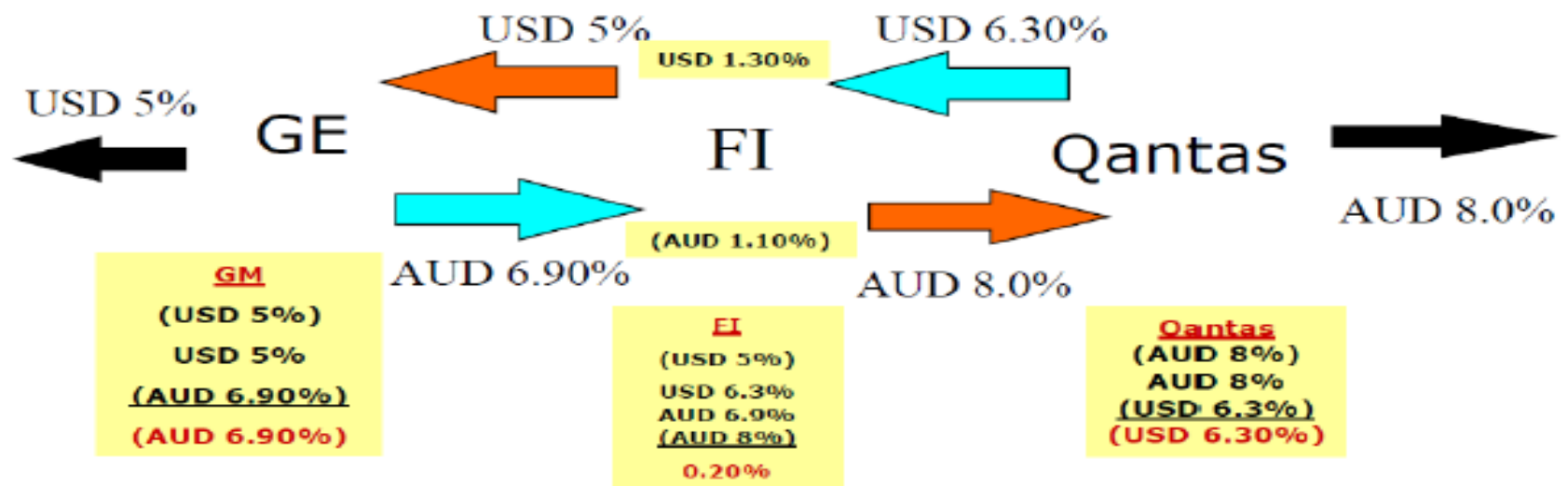
➤ Alternatively:

- The benefit for GM to access Qantas's AUD rate is
 $b = 8\% - 7.60\% = 0.40\%$
- The benefit for Qantas to access GM's USD rate is
 $a = 7\% - 5\% = 2\%$
- The total gain from the swap is
 $a - b = 2\% - 0.4\% = 1.60\%$
- FI wants 0.20% as commission
- Thus each party will benefit by
 $(1.60\% - 0.20\%) / 2 = 0.70\%$

Design of Currency Swaps

- Step 1: GM borrows at USD 5% and Qantas borrows at AUD 8% from outside lenders
- Step 2: GM receives USD 5% and Qantas receives AUD 8% from FI
- Step 3: GM pays AUD (7.6- 0.70)% = 6.90% and Qantas pays USD (7-0.70)% = 6.30% to the FI
- Both parties have improved their borrowing rate by 0.70%.

Design of 0.20% for FI:



Value of Currency Swaps

- Like interest rate swaps, currency swaps can be valued as
 - ***The difference between the values of two bonds***
 - A portfolio of forward rate agreements (FRAs)
- Valuation in terms of bond prices
 - The spot exchange rate S_0 is required
- **Example:** A currency swap where the foreign currency F is paid, and the domestic currency D is received
 - It is a short position in a foreign currency bond and a long position in a domestic currency bond thus

$$V_S = B_D - S_0 B_F$$

Value of Currency Swaps

Example: 10% on £20 million for 6% on \$30 million with 15 months remaining

- Receive \$ and pay £ annually
- Swap discount rate is 4% p.a. in \$ and 7% p.a. in £ with annual compounding for all maturities
- Exchange rate is 1 £= \$1.85
- Thus, \$ payments = \$30 x 6% = \$1.8 million

$$B_D = \frac{1.8}{\frac{1.04^{\frac{3}{12}}}{3}} + \frac{31.8}{\frac{1.04^{\frac{15}{12}}}{15}} = \$32.061 \text{ million}$$

- £payment = £20 x 10% = £2 million

$$B_F = \frac{2}{\frac{1.07^{\frac{3}{12}}}{3}} + \frac{22}{\frac{1.07^{\frac{15}{12}}}{15}} = £22.182 \text{ million}$$

- Value of the swap to the party paying £:

$$V_S = B_D - S_0 B_F = \$32.061 - 1.85 * 22.182 = -\$8.976 \text{ million}$$

- Value of the swap to the party paying \$: $V_S = 8.976$ million



Currency Swaps: What for?

The rationale for currency swaps

- ***Low-transaction-cost*** instrument for changing the currency of denomination of debt financing
- ***Differences in credit risk*** analysis allows for lowering the cost of debt using swaps
- Used to be driven by ***regulatory constraints and tax arbitrage*** opportunities
 - Regulations on types of debt instruments institutions can hold as well as accounting/tax differences have also contributed to the popularity
- Very popular due to the ***integration*** of the world's international financial system

Swap and Forward

Why swaps and not forwards?

- Liquidity
 - Long-dated forward markets are less liquid (or non-existent)
- Trading cost
 - Large bid-ask spreads beyond a maturity of 1 year
- Term
 - FX swaps can provide long-term financing in foreign currencies
- Hedging
 - Associated cash flows of swaps are just like bonds
 - They can easily be hedged in the bond market

Credit Risk

Credit (Default) Risk

- Credit risk is arising from the possibility of a default by the counterparties
- A swap is worth zero initially (similar to a forward contract).
- At a future time, its value may be either positive or negative
- The FIs are exposed to credit risk only when the swap value is positive for the FIs
 - When the swap value to the FI is negative then the counterparty will sell the contract to profit from the positive value of the contract
- Potential losses from defaults on swaps are much less than losses from defaults on loans with the same principle
- At default ***currency swaps generate greater losses than interest rate swaps***

Credit Derivatives

Credit derivatives:

- The payoffs depend on the creditworthiness of one or more companies or countries
- To trade credit risk in much the same way that they trade market risks
- Long by banks and short by insurance companies
- Examples
 - Credit default swaps (CDS);
 - CDS forwards and options
 - Basket CDS;
 - Collateralized debt obligation

Credit Default Swaps (CDS)

Credit Default Swap (CDS)

- Allow companies to hedge credit risk (in the same way of hedging market risk)
- Like an insurance contract that pays off if a particular company or country defaults

CDS spread:

- An insurance premium the buyer of credit protection pays to the seller for default

Example:

- The notation principal of a CDS is \$100 million and the CDS spread for a 5-year deal is 120 basis points.
- The CDS premium: $\$100\text{m} \times 1.2\% = \1.2 m
- Nothing is received if no default during the 5 years
- If default and the bonds are worth 40%, the seller pays 60%, bringing the value of the bonds back to \$100 m.



Appendix: Overnight Indexed Swap (OIS)

- An **Overnight Indexed Swap (OIS)** is a swap where a fixed rate is exchanged for the geometric average of the overnight rates.
- The OIS allows overnight borrowing or lending to be swapped for borrowing or lending **at a fixed rate - the OIS rate**.
- LIBOR-OIS spread, the excess of the 3-month LIBOR rate over the 3-month OIS rate, is used as a measure of stress in financial markets.

3-Month LIBOR-OIS Spread Since the Financial Crisis



Source: Morgan Stanley Research, Bloomberg

Credit Indices

Credit Indices

- Track credit default swap spreads in the credit market, measuring the average CDS spreads.
- Two important indices
 - CDX NA IG, a portfolio of 125 investment-grade companies in North America
 - iTraxx Europe, a portfolio of 125 investment-grade companies in Europe

Example:

- A 5-year CDX NA IG is quoted by 65-66bps per dollar of the notational principal
- You can buy CDS protection on all 125 companies in the index for 66bps
- To protect \$800,000 on each company, the total cost: $\$800,000 \times 0.0066 \times 125 = \$660,000$ per year
- When one company defaults, you receive the usual CDS payment and the annual payment is reduced by $\$660,000/125 = \$5,280$.

Credit Forwards and Options

Credit Forwards

- The obligation to buy/sell a particular CDS on a company at time T ; if the company defaults before T , the contract ceases to exist
- A bank sells a 5-year CDS of a company forward starting in 1 year. If the company defaults within 1 year, the forward contract would cease to exist

Credit Options

- A CDS option is an option to buy/sell a particular CDS at a time T
- A CDS call to buy 5-year protection on a company starting in 1 year for 280bps will be exercised only when the 5-year CDS spread in 1 year turns out to be more than 280bps.

Basket CDS

Basket CDS

- With a number of reference companies
- An add-up basket CDS provides a payoff when any of the reference companies default.
- A first-to-default CDS provides a payoff when the first default occurs
- A kth-to-default CDS provides a payoff when the kth default occurs

Collateralized Debt Obligations (CDO)

CDO

- An **Asset-backed security (ABS)** where the underlying assets are bonds
- A **waterfall** is defined as the interest and principal payments on the bonds to ensure that, if one **tranche** is more **senior** than another, it is more likely to receive promised interest payments and repayments of principal.

Caps and Floors

➤ Interest rate caps and floors

- Interest rate caps and floors are **portfolios of call and put** options on interest rates.
- A **cap (call option)** can insure a floating rate borrow to be below certain given level;
- A **floor (put option)** can ensure that returns of bond investment to be above certain level.

Liquidity Risk

- If a bank funds long term assets with short term liabilities such as commercial paper, it can use FRAs, futures, and swaps to hedge its interest rate exposure
- But it still has a liquidity exposure.
- It may find it impossible to roll over the commercial paper if the market loses confidence in the bank
- ***Northern Rock*** is an example of this type of liquidity problem